Practical - 8

Practical Title: MySQL Schema Design, ER Model, PDO Connectivity, and Prepared Statements.

Design the database for StudentHub with students, events and registrations. Create PHP database connection using MYSQL statements.

A. Theory

1. **Is the schema normalized properly?**

Yes, the database is normalized by storing students, events, and registrations in separate tables to reduce data redundancy and maintain data consistency.

2. **Are primary keys and foreign keys correctly defined?**

Yes, primary keys uniquely identify each record, while foreign keys in the registrations table link students and events and maintain referential integrity..

3. **Is database connection handled securely?**

Yes, PDO is used with exception handling, UTF-8 encoding, and prepared statements to handle database errors and help prevent SQL injection.

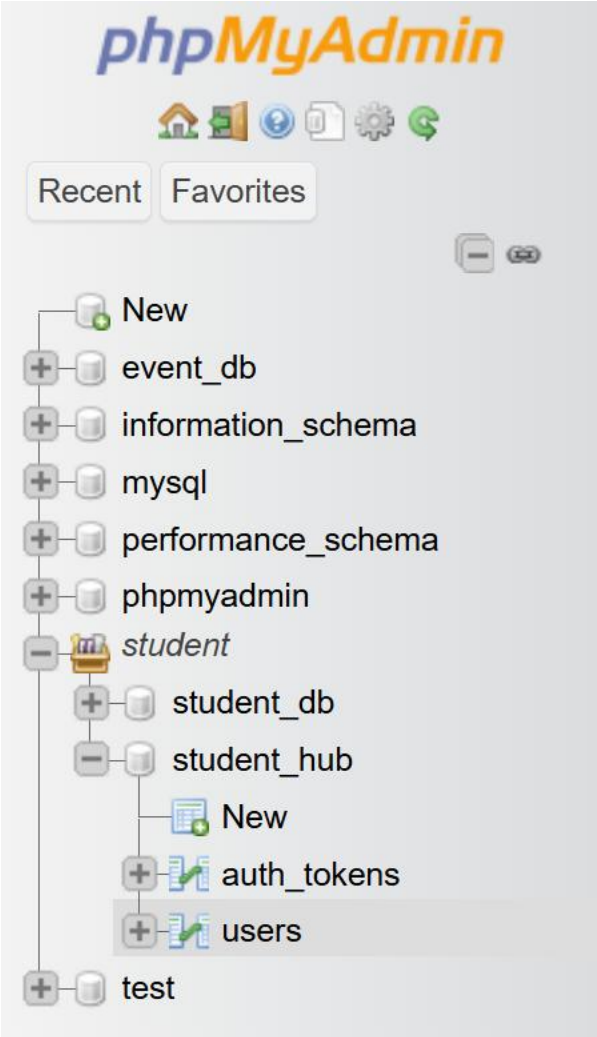
Implementation/ Output/Result

Tags:

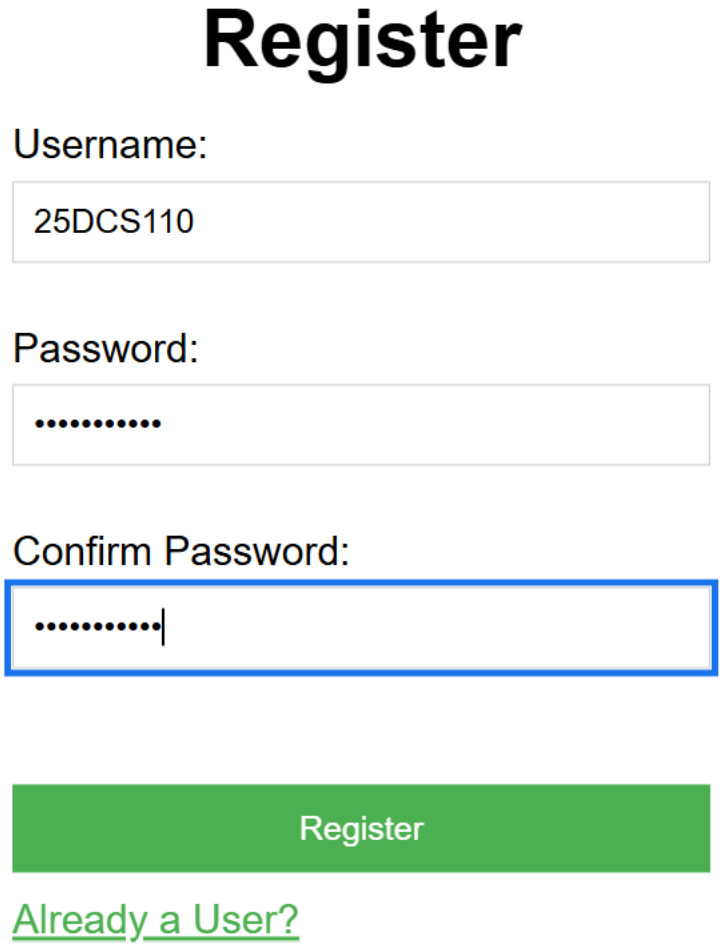
- Created a MySQL database named studenthub with students and registrations tables.
- Defined primary keys and foreign keys to establish relationships between tables.
- Established a database connection using PHP PDO.
- Used prepared statements to securely insert and retrieve student details.

Output Screenshots:

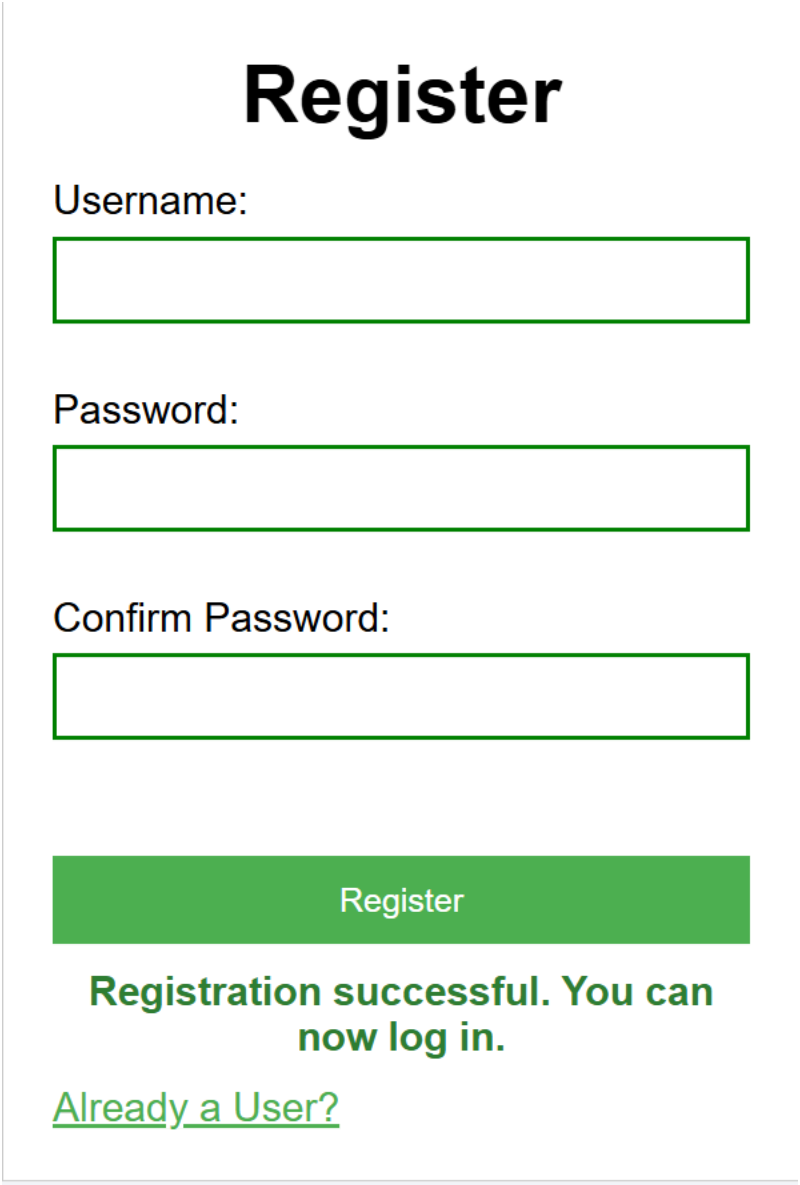
Database structure:



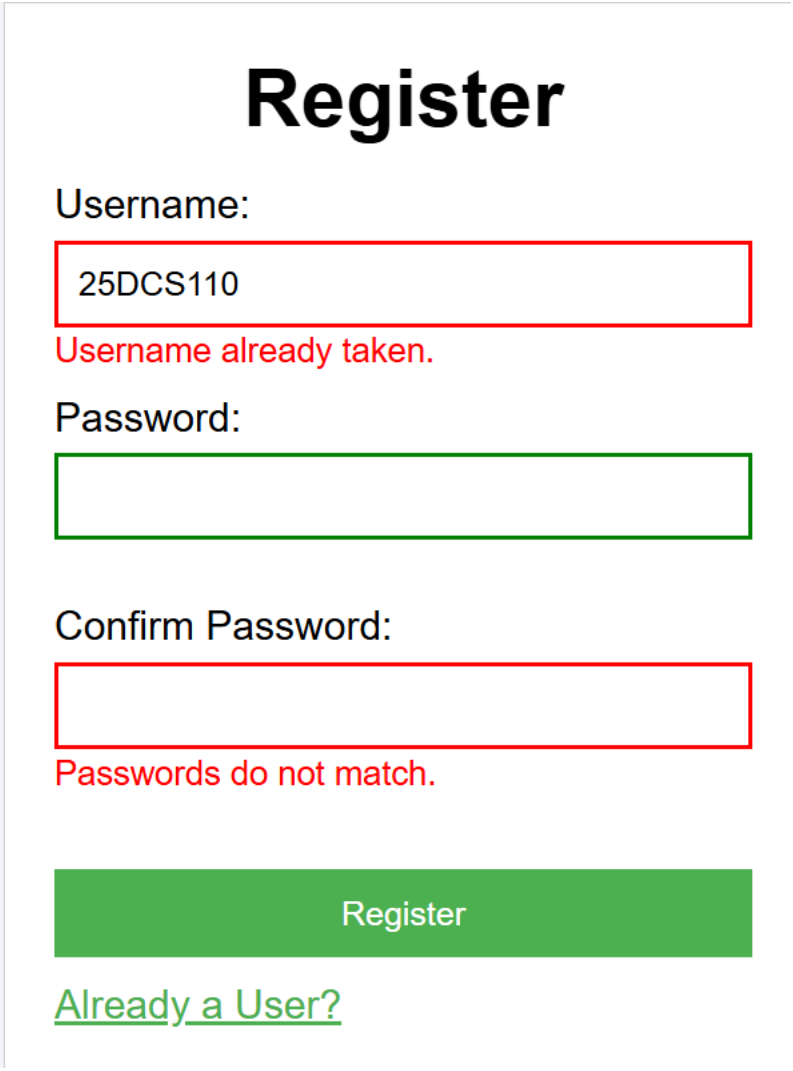
New Student Registration:



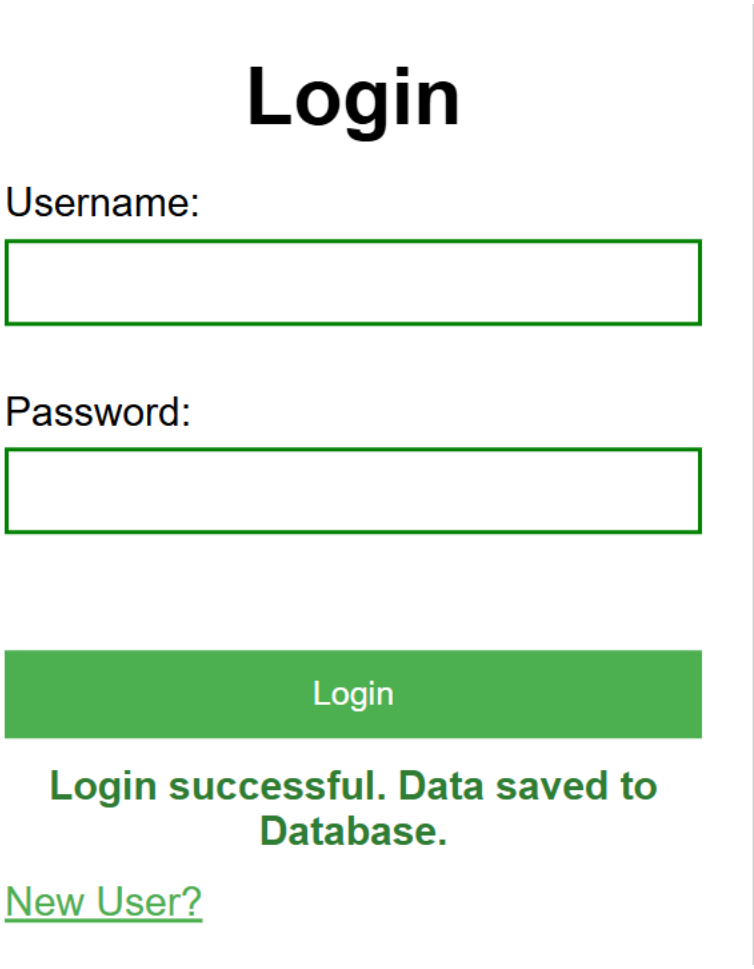
After Clicking Register Button:



Registration of Existing Student:



Successful Login for Register Student:



4. GitHub Repository Link:

https://github.com/HimeshMale/Student-Hub-Portal.git

5. Challenges Faced:

It was not too difficult but I faced challenge in creating a php file and retrive data from student database table into events table and in PHP page.

6. Learning Outcomes:

Learned to design a normalized MySQL database, establish a secure PDO connection, and use prepared statements to insert and retrieve data securely.

Faculty Remarks: